

FEASIBILITY REPORT

Centralized College Society Management Portal

Course: Software Engineering

College Name: Thapar Institute of Engineering & Technology

Team Members:

Tulsi Relia (102303984)

Shambhavi (102303987)

Khushi Goyal (102303993)

Tanmay Kapoor (102303966)

Date: 20-08-2025

1. Introduction

SocietyConnect is a centralized college event management and recommendation platform designed to simplify the interaction between student societies and students. It automates event categorization using AI-powered tag prediction and delivers personalized event recommendations to students based on their selected interests. The system makes event discovery easier, improves participation, and reduces the communication gap between societies and the student community.

2. Problem Statement / Need for the System

In most colleges, including Thapar Institute, the communication of events is scattered across:

- Instagram pages
- WhatsApp groups
- College websites
- Individual society announcements

This leads to several issues:

- Students miss relevant events because announcements are not centralized.
- Societies spend time manually categorizing and tagging events.
- Students receive irrelevant event updates, causing low participation.
- There is no platform where students can filter events by interest.

Need:

A centralized platform where societies can easily post events and students can instantly discover events aligned with their interests. Automation was necessary to make it easy for admins and personalized for students.

3. Objectives

- To build a user-friendly platform for society admins to create and manage events.
- To implement AI-based tag prediction so societies do not need to manually classify events.
- To allow students to select their interests and receive personalized event recommendations.
- To maintain a lightweight, responsive UI for students and societies.
- To integrate secure authentication for role-based login (student / society).
- To store event data using a scalable cloud database (Supabase).

4. Feasibility Study

A. Technical Feasibility

The system uses:

- Frontend: React + Vite
- Backend: FastAPI (Python)
- Database: Supabase (PostgreSQL + Auth)
- AI Engine: Google Gemini 2.5 Flash

All technologies are modern, well-supported, easy to integrate, and free for individual academic use. The architecture supports future scaling and mobile-app integration.

B. Economic Feasibility

The project uses free tiers of:

- Render (backend hosting)
- Netlify/Vercel/Render (frontend hosting)
- Supabase free tier with authentication + database

- Gemini free tier for AI tag classification

No hardware cost. Suitable for student projects and real deployment.

C. Operational Feasibility

- Simple UI (Login → Dashboard → Events)
- Roles clearly separated: student vs. society
- AI auto-tagging reduces manual workload
- Students get highly relevant events, improving participation

Overall, easy to operate and maintain.

5. System Overview

A. Society Admin Panel

Society admins can:

- Log in securely
- Create new events
- Let AI generate tags based on event description
- View all previously added events
- Delete events

B. Student Panel

Students can:

- Log in using Thapar email
- Choose interests (Tech, Art, Coding, etc.)
- View events recommended based on selected interests
- See matched tags for transparency

C. AI Tag Prediction

Event descriptions are sent to Gemini API, which predicts the best tag categories from a fixed list such as:

- AI/ML
- Coding
- Robotics
- Cultural
- Sports
- Music
- Marketing

...and more.

If the model is uncertain, fallback logic ensures tags are consistent.

6. Technology Used

- React (Frontend) – Fast, modular UI framework
- FastAPI (Backend) – High-performance API layer
- Supabase – Authentication + Realtime DB
- Google Gemini – Tag prediction model
- Render – Deployment for backend
- CSS – Custom styling

7. Implementation Details

A. AI Tag Prediction

The backend sends the event description to Gemini using a structured prompt. The model returns the most probable tags. The system cleans output, validates tags, and ensures fallback categories for reliability.

B. Interest-Based Event Matching

Students choose their interests once.

Events are filtered by checking:

$\text{event.tags} \cap \text{student.interests} \neq \emptyset$

This ensures students see only relevant events.

C. Authentication & Roles

- Only @thapar.edu emails allowed
- Role stored in profiles table in Supabase
- Login checks for role mismatch (student cannot log in as society)

8. User Evaluation

A survey was conducted, and here are key insights:

What users liked most:

- “AI auto-tagging makes event creation fast and smart.”
- “The platform shows only events I actually care about.”
- “Tag prediction is accurate even for short descriptions.”
- “Interest-based event recommendations feel personalized.”

★ Observed outcomes:

- Students find relevant events 2–3× faster
- Societies reduce manual tagging effort completely
- Navigation is smooth and requires no training
- Email verification improves platform security

★ Summary:

Users found SocietyConnect extremely helpful, intuitive, and time-saving.

9. Advantages of SocietyConnect

- Centralized event discovery
- Smart AI-driven tag classification
- Personalized recommendations
- User role-based authentication
- Clean, modern UI
- Very helpful for new students to find relevant clubs/events

10. Limitations

- AI tag prediction depends on internet and API availability
- No admin analytics dashboard yet
- Announcements are not stored in the database (local only)
- Only supports Thapar emails currently

11. Future Enhancements

- Add reminder notifications
- Add society verification using official credentials
- Profile dashboard with badges and event history
- Real-time updates when new events are posted
- Mobile app version
- Push notifications
- Event RSVP / ticketing
- Analytics dashboard for societies

12. Conclusion

SocietyConnect successfully bridges the communication gap between societies and students. Its AI-driven tag prediction reduces workload for societies, while interest-based event recommendations create a personalized experience for students. The system is technically feasible, user-friendly, secure, and scalable. It can be adopted as an official college event portal with minimal modifications.